



Data Integration & Applications

Data Integration & Applications (ST2DLDI)

Training Plan

- Data Types
- Storage solutions
- Data management evolution
- Relational Databases
- Modern storage architectures
- Data Quality and Governance
- *Case Study*

Learning Objectives

- Understand principles, architectures, and challenges of data integration
- Apply cleaning, transformation, and integration techniques
- Prepare unified datasets for analytics and ML

Data Types

- Organizations collect vast amounts of data from diverse sources:
 - **Structured Data** (databases, spreadsheets)
 - **Semi-Structured Data** (JSON, XML)
 - **Unstructured Data** (images, videos, free text)
- Goal : Efficient storage and analysis solutions for diverse data types



Practical Exercise #1

Part 1 : Data Exploration

- In Kaggle website, select a dataset for each data type:
 - **Structured**
 - **Semi-structured**
 - **Unstructured**

Part 2 : Data Storage Design

- Outline how each dataset would be stored in a Data Warehouse vs. a Data Lakehouse.
- Identify any transformations needed for analysis in each architecture.

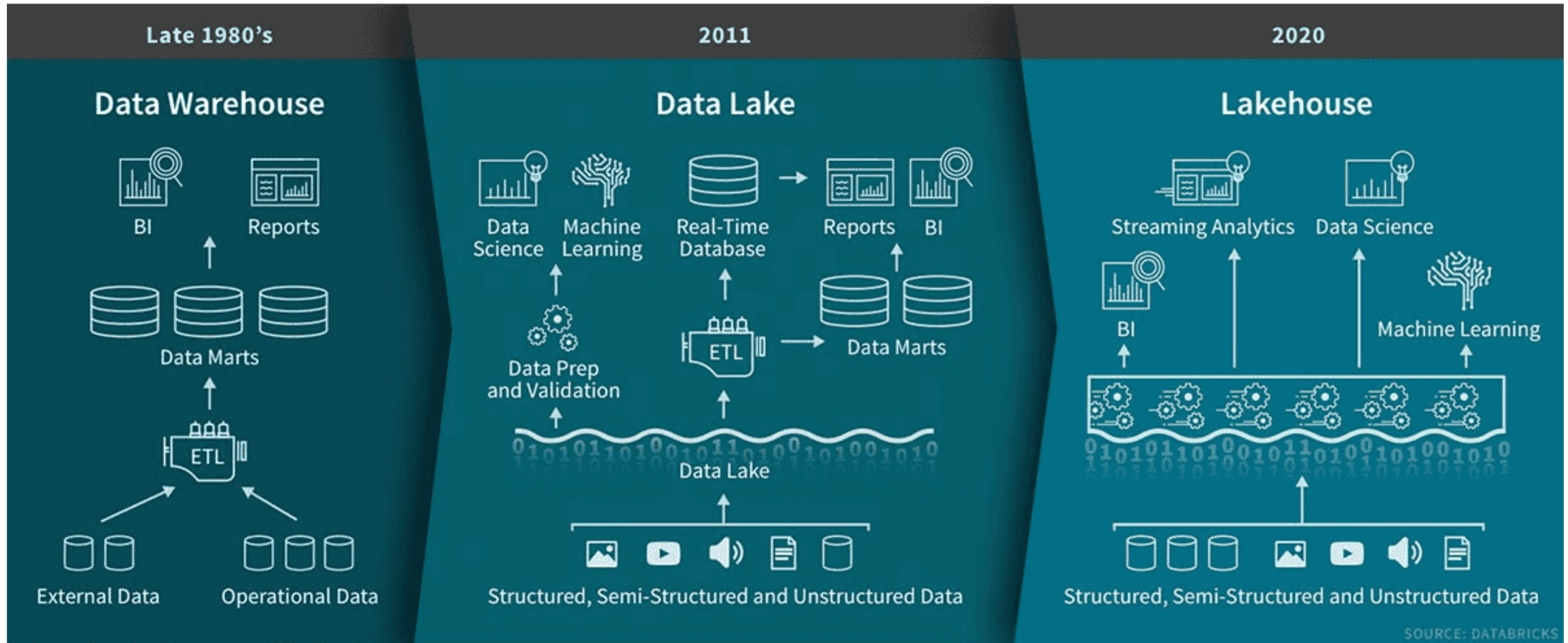
Part 3 : Comparison Analysis

- Create a table of pros/cons for storing and analyzing each data type in a Data Warehouse vs. Lakehouse.

Storage solutions

- **Data Warehouse:** Traditionally built for **structured data** and optimized for fast, structured analysis using rigid schemas.
- **Data Lake:** Stores vast amounts of raw, unprocessed data in any format, offering flexibility for big data analysis and machine learning
- **Data Lakehouse:** Combines the strengths of **Data Warehouses** and **Data Lakes**

Storage solutions



Source : DataBricks

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What is Data Management ?

- **Data management** involves the strategies, frameworks, and tools used to consistently access and distribute data across different formats within a company.
- ➔ Meet all data needs for various applications, business operations, and use cases throughout the organization.



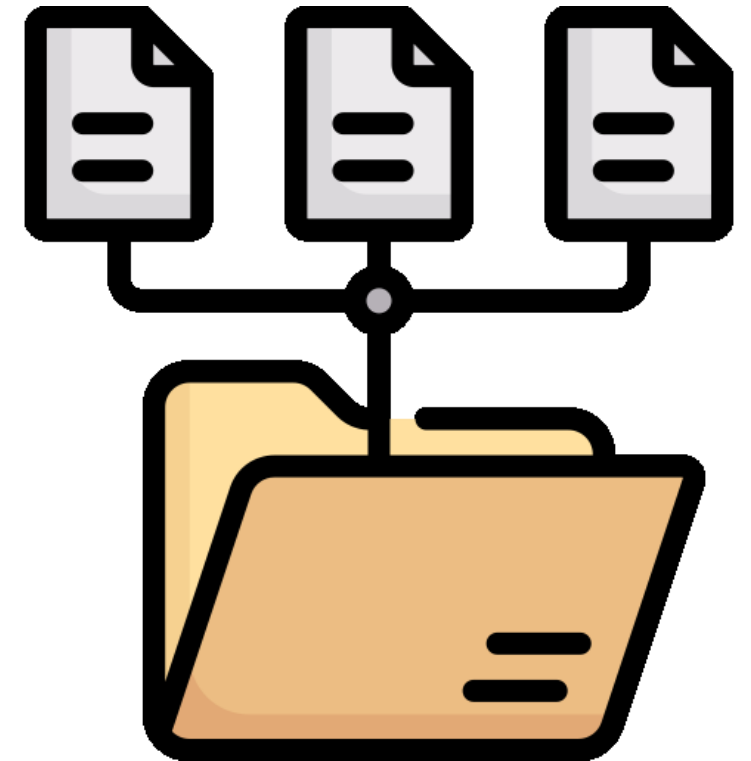
Data Management : evolution over time

- Early relational databases
- Department silos
- Explosion of data
- Need for scalable systems



Relational Databases

- SQL
- Structured data
- Strengths/Limits



Practical Exercise #2

Data Modeling with structured data

Prerequisite

- Open the file : adventureworks Sales.xlsx

Part 1: Data Analysis

- **Objective:** Explore and understand the structure and content of the **AdventureWorks Sales.xlsx** file.
 - **Task:** Review each table to understand its purpose and the types of data it contains.
 - **Questions :**
 - 1) What is the role of each table in the sales process?
 - 2) Which fields seem to uniquely identify records, and which fields might link tables together?

Practical Exercise #2

Data Modeling with structured data

Part 2: Key Identification

- In each table, identify:
 - **Primary Key:** The unique identifier for each record.
 - **Foreign Keys:** Keys that link to other tables, establishing relationships.

Part 3: Entity-Relationship Diagram (ERD)

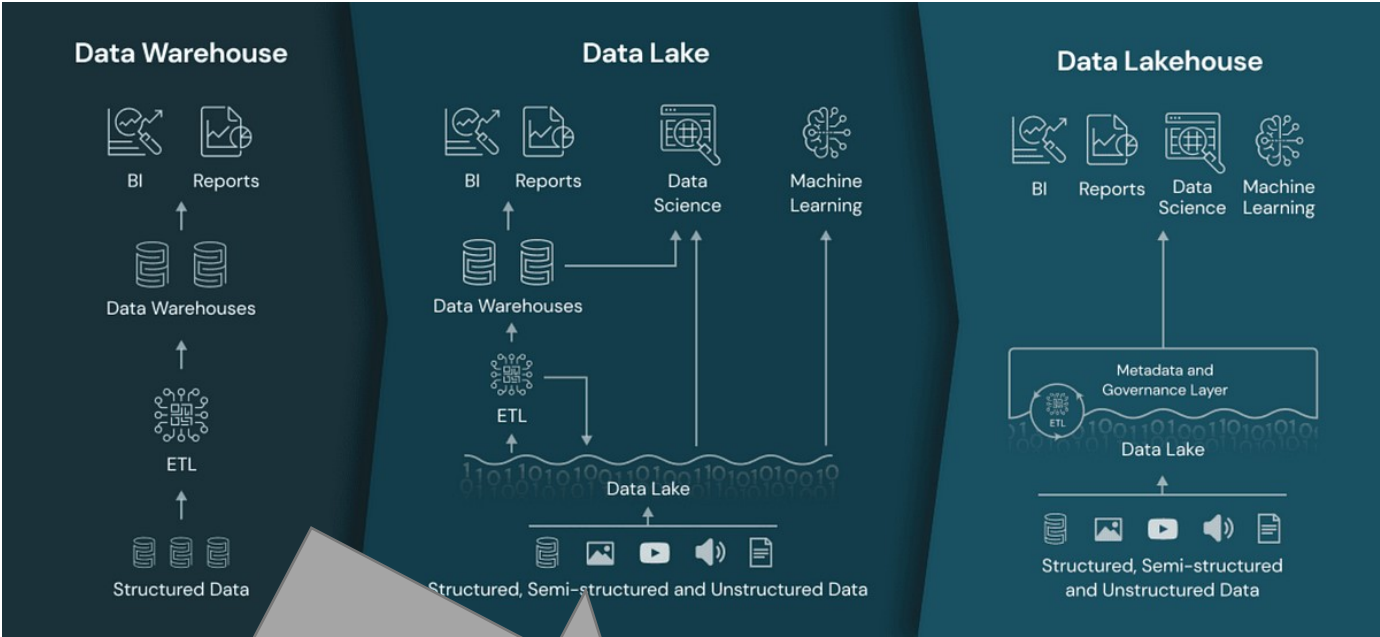
- Using **draw.io**, create an ERD that includes:
 - Each table as an entity, listing columns.
 - Underline primary keys and use arrows to indicate foreign keys linking tables.

Part 4: Relationship Analysis

- Describe the relationships between tables as One-to-Many, Many-to-One or Many-to-Many.

Modern storage architectures

Source : [Databricks](#)

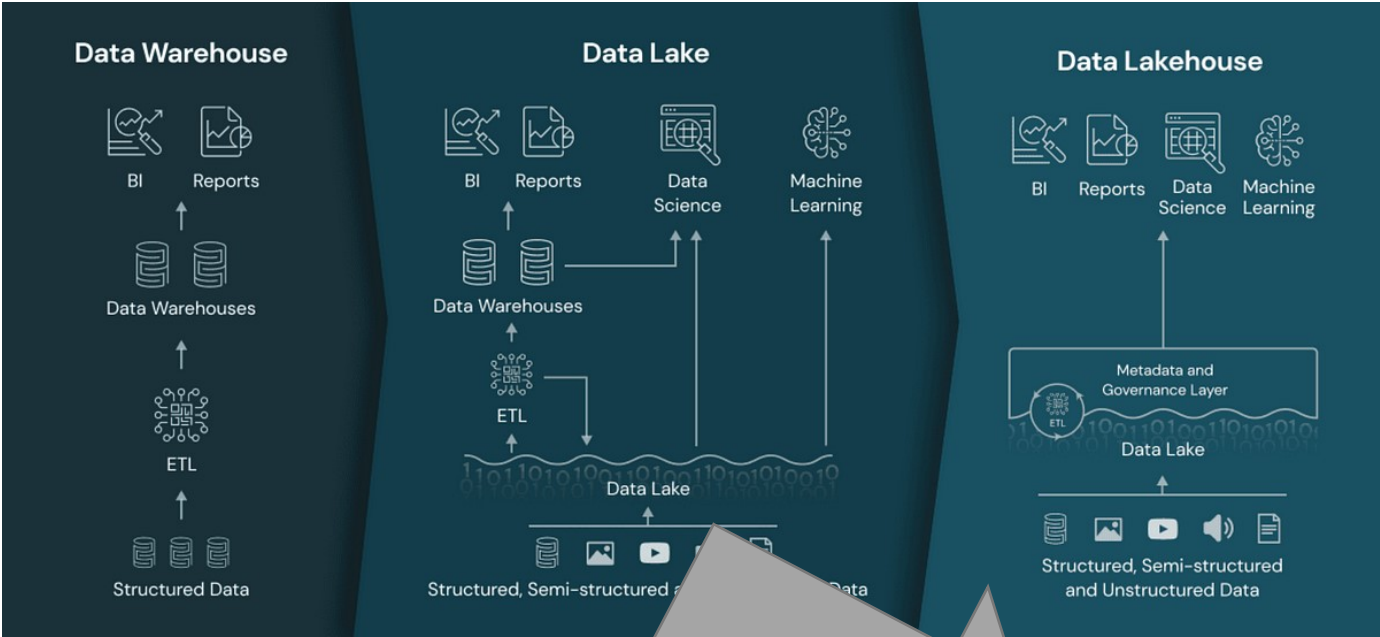


Data Warehouses

PROS	CONS
Strong governance	Rigid schema
High performance for structured data	Poor support for semi/unstructured data
Security & compliance	Expensive at scale
Ideal for BI	

Modern storage architectures

Source : [Databricks](#)

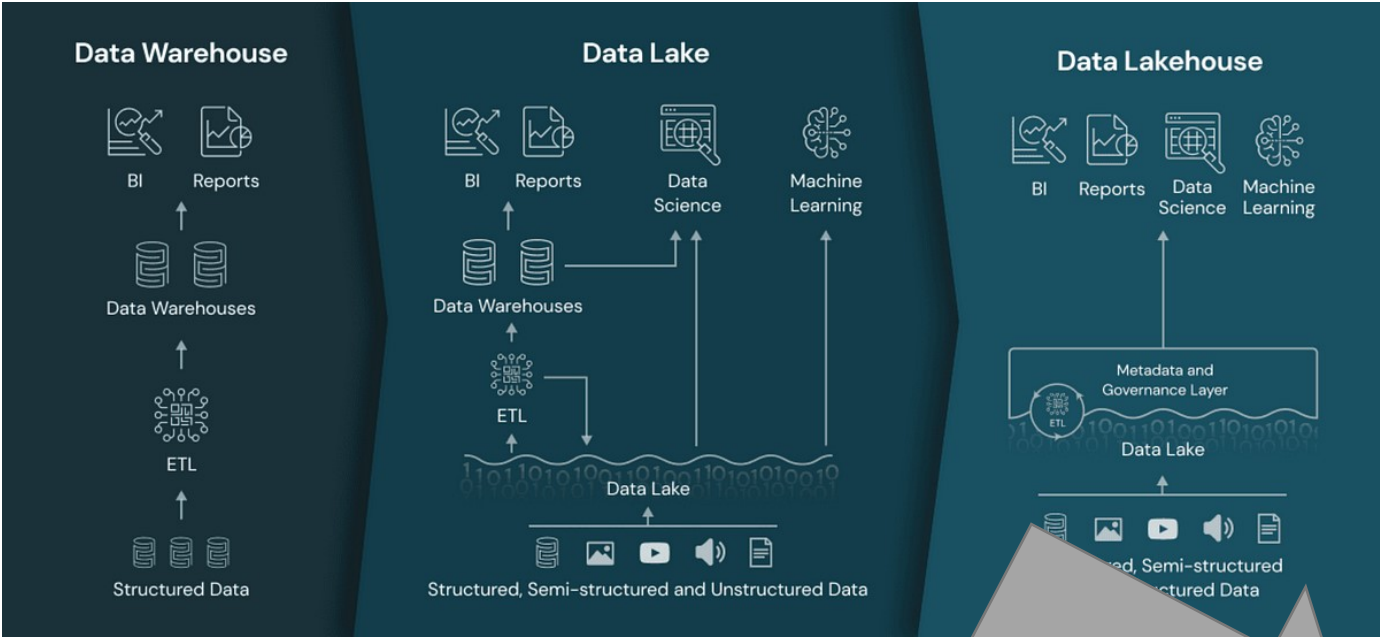


Data Lake

PROS	CONS
Handles all data types	<u>Weak governance</u>
Cheap storage	Harder analytics
Flexible ingestion	

Modern storage architectures

Source : [Databricks](#)

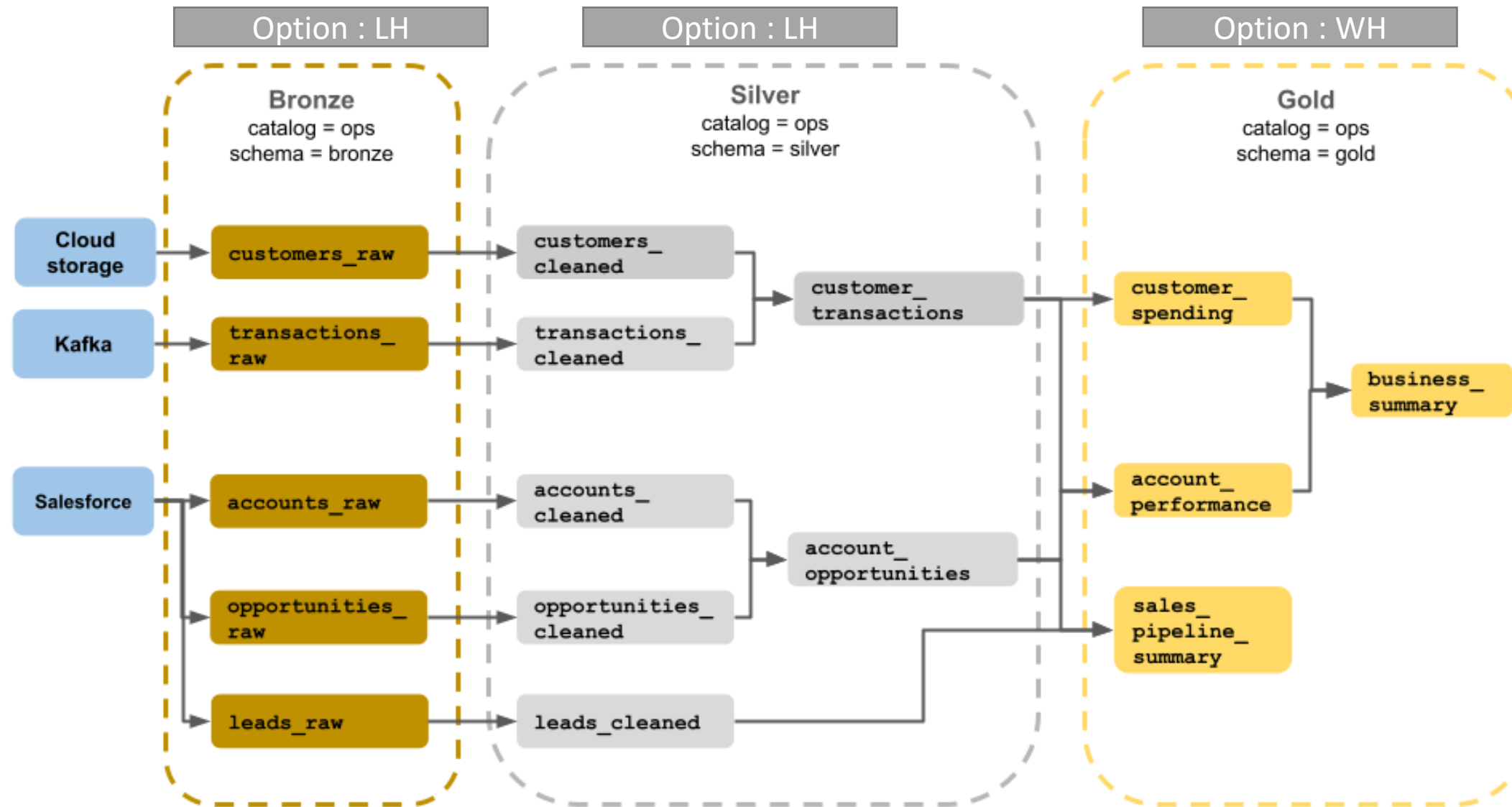


Data Lakehouse

PROS	CONS
Combines flexibility of lakes + performance of warehouses	More complex to implement
Handles structured + semi + unstructured (as Data Lakes)	Requires modern tooling
Supports BI + ML	
<u>Modern governance (Delta/Parquet)</u>	

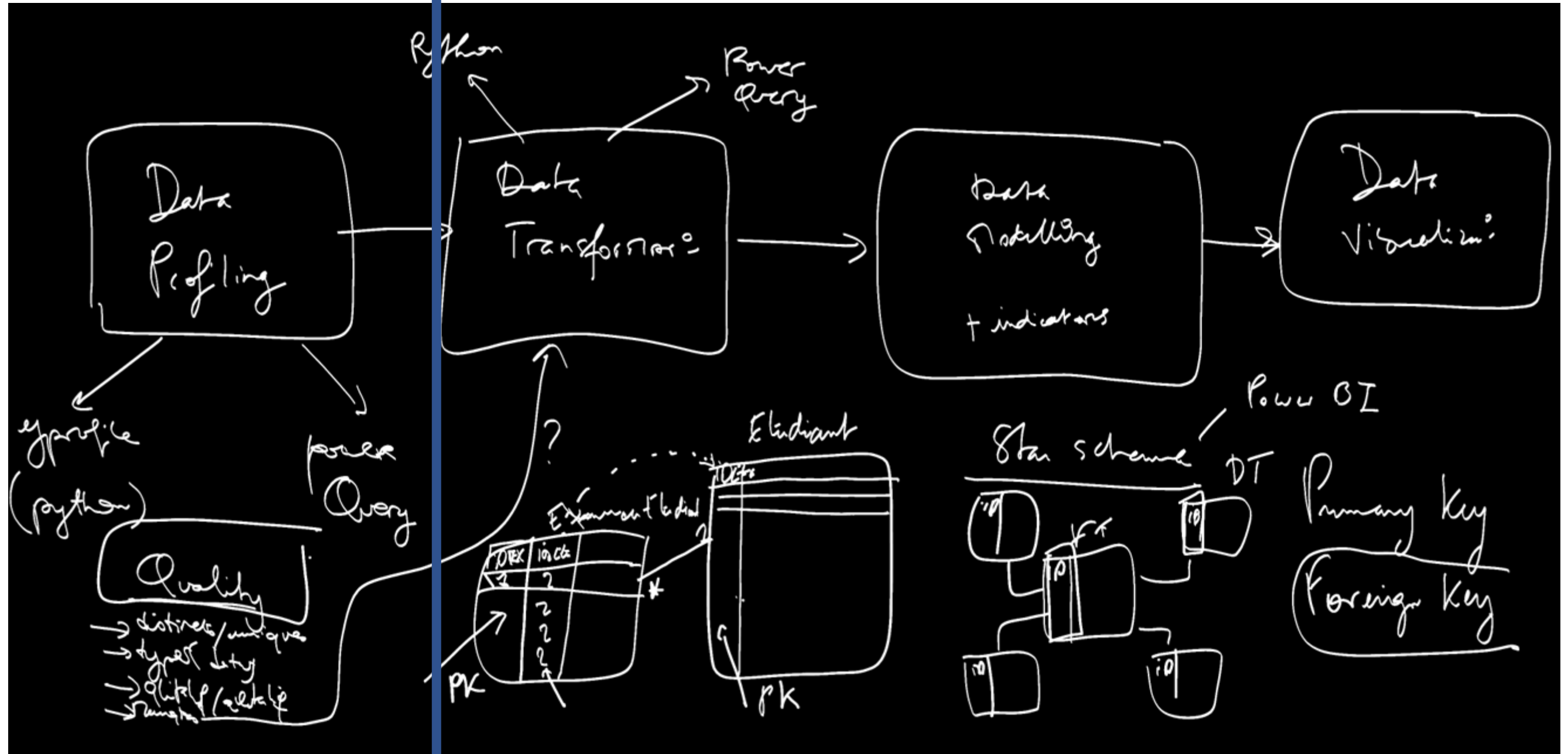
Modern storage architectures

The Medallion Architecture



Source : MSLearn

Data Profiling and Quality



Data Profiling and Quality

- Accuracy
- Validity
- Completeness
- Uniqueness
- Timeliness

Data Profiling and Quality

Demos (Python yprofile/Power Query)

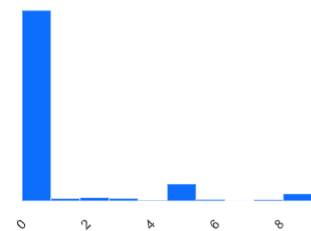
Dataset statistics		Variable types	
Number of variables	47	Numeric	21
Number of observations	121309	Categorical	14
Missing cells	402377	DateTime	1
Missing cells (%)	7.1%	Text	8
Duplicate rows	3	Boolean	3
Duplicate rows (%)	< 0.1%		
Total size in memory	162.9 MiB		
Average record size in memory	1.4 KiB		

infra

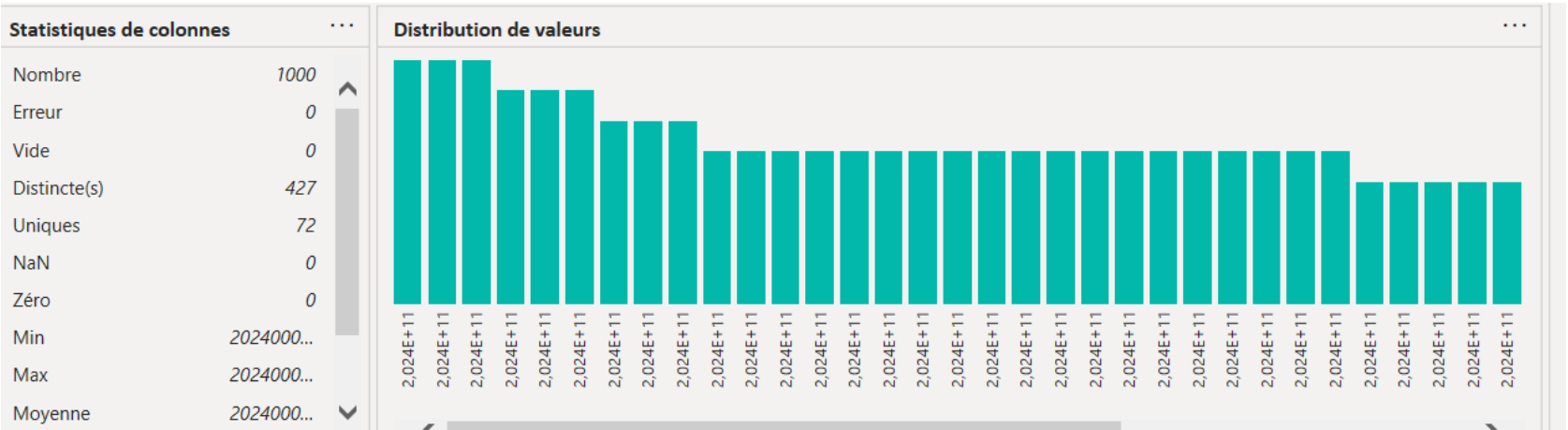
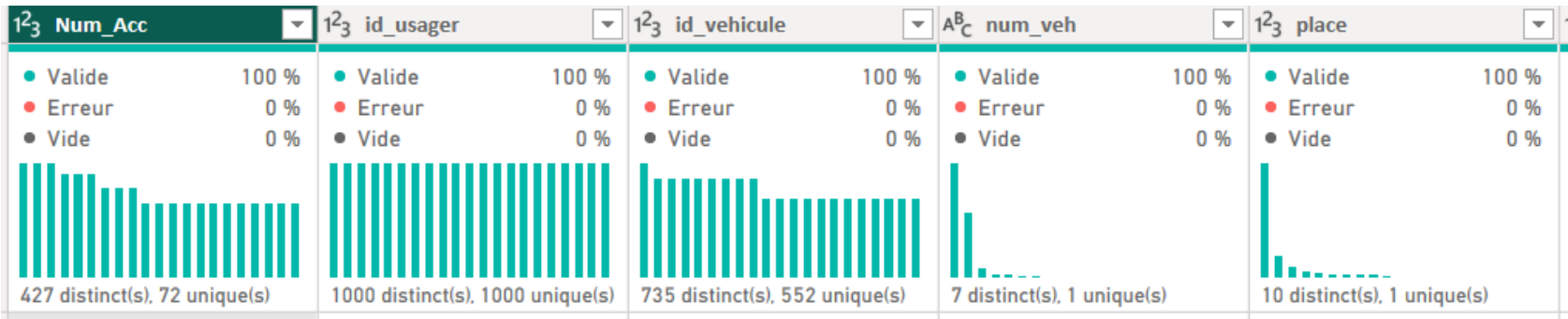
Real number (ℝ)

MissingZeros

Distinct	10	Minimum	0
Distinct (%)	< 0.1%	Maximum	9
Missing	1337	Zeros	100505
Missing (%)	1.1%	Zeros (%)	82.9%
Infinite	0	Negative	0
Infinite (%)	0.0%	Negative (%)	0.0%
Mean	0.84678925	Memory size	947.9 KiB



More details

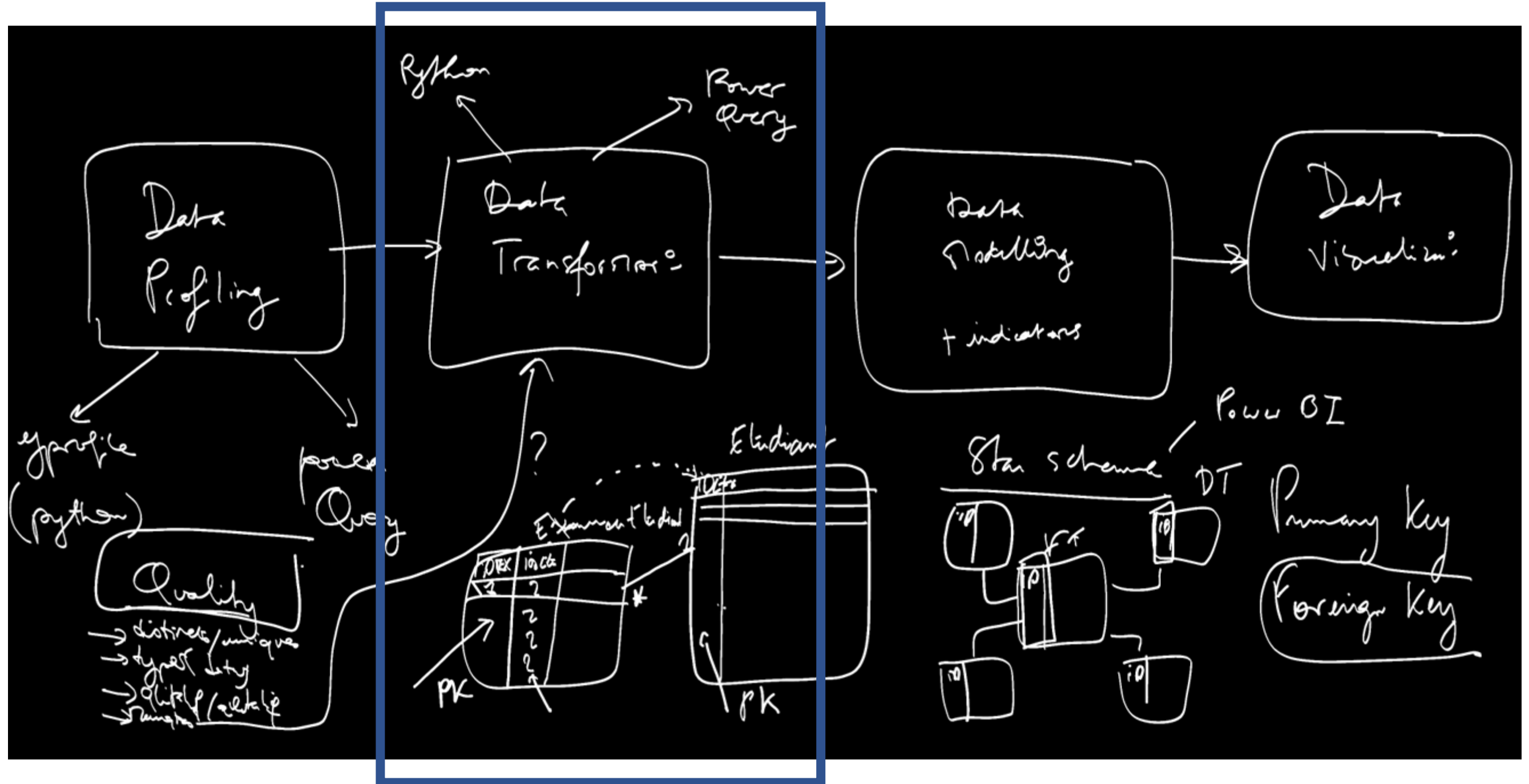


Data Profiling and Quality

Cleaning strategies

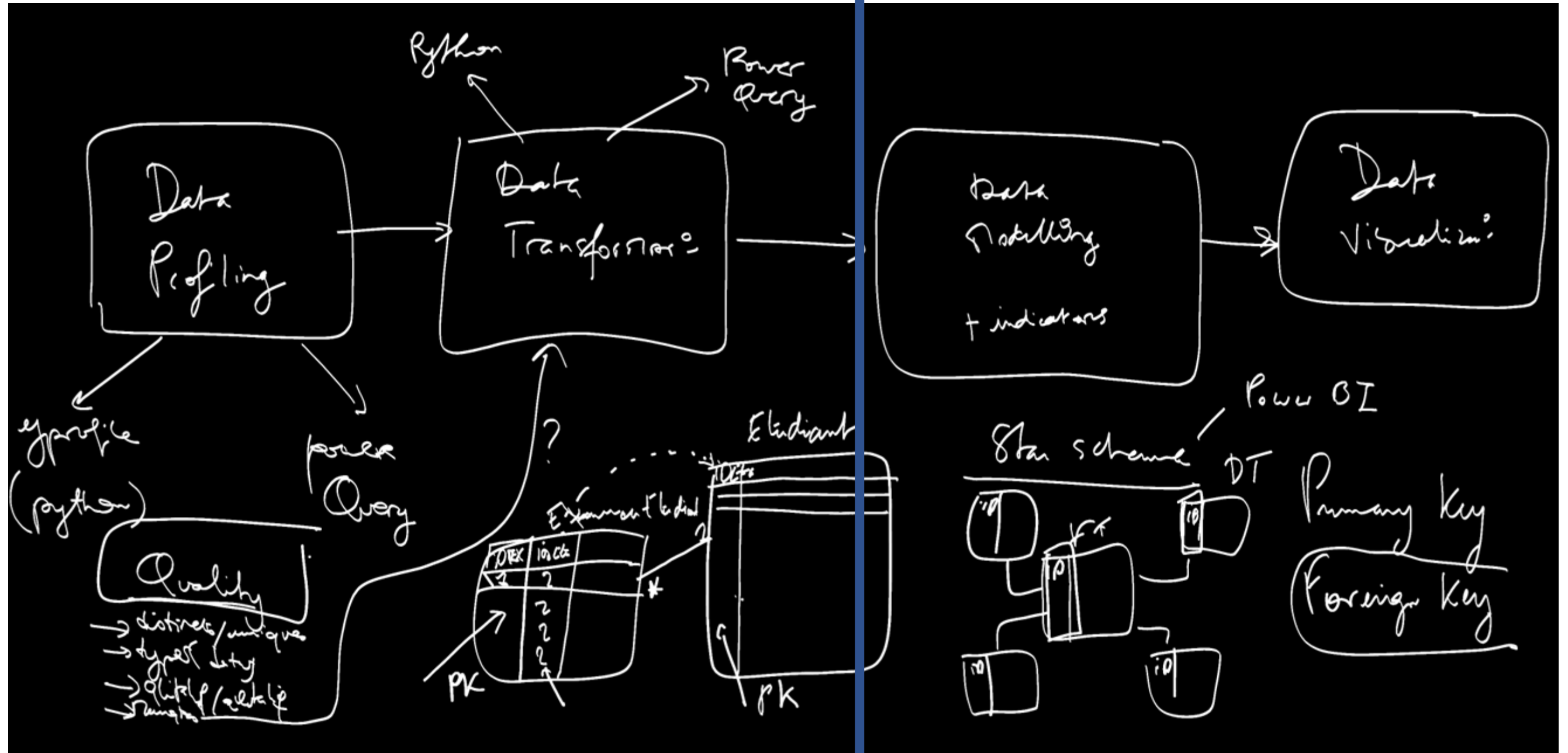
- Mean/median imputation
- Duplicate handling
- Null management
- PK integrity checks

Data Transformation Workflow



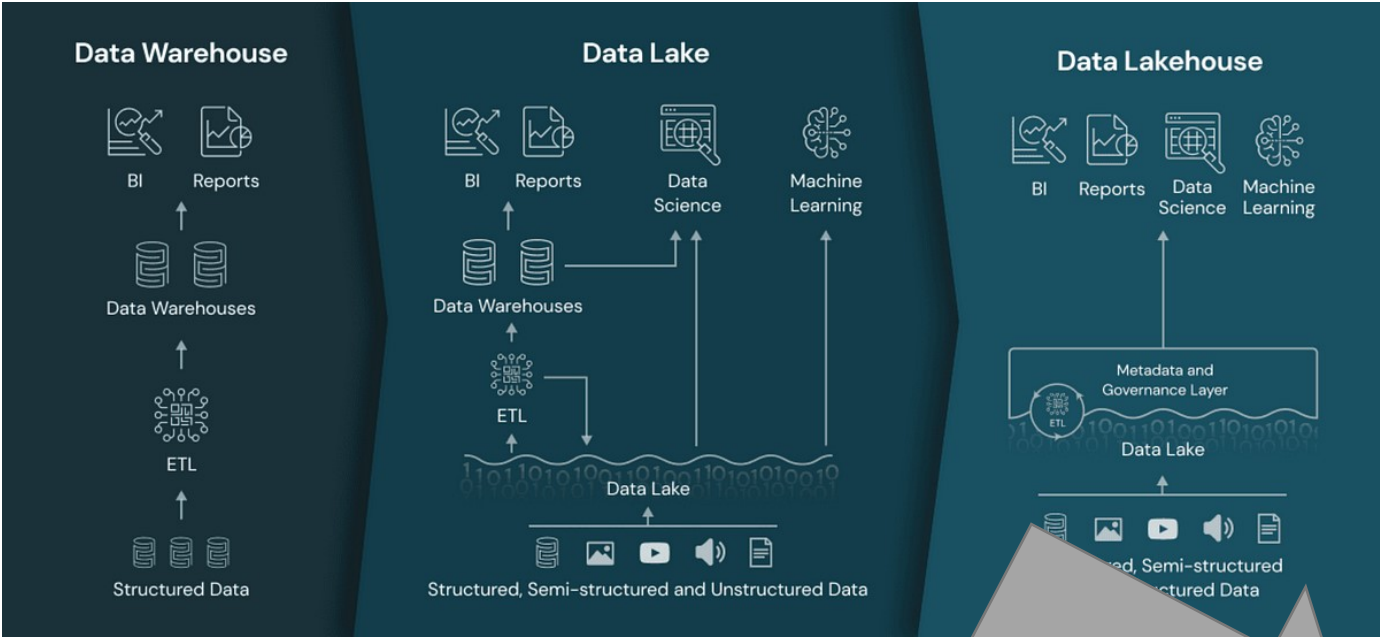
Ex: Python cleaning (null handling, deduplication, filtering)

Data Modeling, enrichment and data visualisation



Modern storage architectures

Source : [Databricks](#)



Data Lakehouse

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Combines flexibility of lakes + performance of warehouses	More complex to implement
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<u>Modern governance (Delta/Parquet)</u>	



Use case

“Sécurité Routière” dataset from data.gouv.fr

Link

[Jeu de données - Bases de données annuelles des accidents corporels de la circulation routière - Années de 2005 à 2024 | data.gouv.fr](https://www.data.gouv.fr/datasets/bases-de-donnees-annuelles-des-accidents-corporels-de-la-circulation-routiere-annees-de-2005-a-2024?resource_id=36496bab-a042-47bf-b08b-3c7467f2bddf)

Data Documentation

https://www.data.gouv.fr/datasets/bases-de-donnees-annuelles-des-accidents-corporels-de-la-circulation-routiere-annees-de-2005-a-2024?resource_id=36496bab-a042-47bf-b08b-3c7467f2bddf